Code No: 114AG **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year II Semester Examinations, December - 2017
FORMAL LANGUAGES AND AUTOMATA THEORY
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

JNTUH USED 16-12-2017AM

PART- A

(25 Marks)

1.a) Define Transition diagram. [2]
b) Explain about ε-closure(). [3]
c) Write a short note on Derivation Tree. [2]
d) Explain Rightmost Derivation with an example. [3]
e) Define Chomsky Normal Form. [2]
f) Design Push Down Automata for $L = \{a^{2n}b^n \mid n \geq 1\}$. [3]
g) Write a short note on Church's hypothesis. [2]
h) Write a short note on Recursively Enumerable languages. [3]
i) Write about decidability of problems. [2]
j) Define NP-complete problems. [3]

PART-B

(50 Marks)

2.a) Convert the Moore machine to determine residue mod 3 into Mealy machine.
b) Construct the minimum state automata for the following. [5+5]

	0	**1**
→A	**B**	**C**
B	**B**	**C**
C	**B**	**C**
D	**B**	**E**
Ⓔ	**B**	**C**

OR

3.a) Convert the following NFA with ε to equivalent DFA:

	a	**b**	**ε**
→P	**Φ**	**P**	**Q**
Q	**Q**	**Φ**	**R**
Ⓡ	**Q**	**P**	**Φ**

b) Design NFA accepting string with a's and b's such that string containing two consecutive a's or two consecutive b's. [5+5]

4.a) For the following grammar give the leftmost and rightmost derivation for the string '00101'.
$S \rightarrow A1B$
$A \rightarrow 0A / \varepsilon$
$B \rightarrow 0B / 1B / \varepsilon$

b) Prove that the following language $\{a^n b^n \mid n \geq 1\}$ is not a regular. [5+5]

OR

5.a) Write a R.E. for the following DFA:

	a	b
→ P	Q	P
(Q)	Q	P

b) Construct the right linear grammar for the language (0+1)*00(0+1)*. [5+5]

6.a) Convert the following grammar to Greibach Normal Form
$S \rightarrow ABA \mid AB \mid BA \mid AA \mid B$
$A \rightarrow aA \mid a$
$B \rightarrow bB \mid b$

b) Convert the following Context Free Grammar into Chomsky Normal Form
$S \rightarrow aaaaS \mid aaaa$ [5+5]

OR

7.a) Reduce the following grammar such that there are no UNIT productions
$S \rightarrow AA$
$A \rightarrow B \mid BB$
$B \rightarrow abB \mid b \mid bb$

b) Construct CFG for the PDAM = $(\{q_0,q_1\}, \{0,1\}, \{R,Z_0\}, \delta, q_0, Z_0, \Phi)$ and δ is given by
$\delta(q_0,1,Z_0)=(q_0,RZ_0)$
$\delta(q_0,1,R)=(q_0,RR)$
$\delta(q_0,0,R)=(q_1,R)$
$\delta(q_1,0, Z_0)=(q_0, Z_0)$
$\delta(q_0,\varepsilon, Z_0)=(q_0,\varepsilon)$
$\delta(q_1,1,R)=(q_1,\varepsilon)$ [5+5]

8.a) Design a TM to recognize the language $L= \{0^n1^n0^n \mid n \geq 1\}$.

b) Design TM which will recognize strings containing equal number of a's and b's. [5+5]

OR

9.a) Design a TM to accept the language $L= \{w\ c\ w^R \mid w\ \varepsilon\ (a+b)^*\}$.

b) Design a TM to recognize all strings consisting of odd number of 1's. [5+5]

10.a) Give the closure properties of deterministic context free languages.

b) What are undecidable problems? Give example. [5+5]

OR

11.a) Give definitions of P and NP problems.

b) Explain TM halting problem. [5+5]

---ooOoo---

JNTUH USED 16-12-2017 AM